Algorithm

Music Playlist Manager using Linked List | Unit 3

Main Driver Algorithm

1. Start.
2. Initialize five empty structures: Playlist (SLL), History (Stack), UpNext (Queue), NavigablePlaylist (DLL), RepeatPlaylist (Circular LL).
3. Display the menu of operations to the user.
4. Read the user's choice.
5. If choice equals Exit (0), go to Step 7.
6. Otherwise, call the function for the chosen operation on the matching structure, then return to Step 3.
7. Free all dynamically allocated nodes and stop.

Add Song (Singly Linked List)

1. Create a new node with the given song name; set its next pointer to NULL.
2. If the list is empty (head == NULL), set head to the new node and stop.
3. Otherwise, traverse from head until a node whose next pointer is NULL is found.
4. Set that node's next pointer to the new node.

Push / Pop (Stack using Linked List: History)

1. Push: create a new node, set its next pointer to the current top, then set top to the new node.
2. Pop: if top is NULL, report empty history and stop.
3. Otherwise, store top in a temporary pointer, move top to top->next, and delete the temporary node.

Enqueue / Dequeue (Queue using Linked List: Up Next)

1. Enqueue: create a new node; if front is NULL, set front = rear = new node.
2. Otherwise, set rear->next = new node and then rear = new node.
3. Dequeue: if front is NULL, report empty queue and stop.
4. Otherwise, store front in a temporary pointer, move front to front->next, and delete the temporary node.

Next / Previous (Doubly Linked List: Navigation)

1. Add song: create a node, go to the last node, set last->next = new node and new node->prev = last.
2. Next: if current->next is not NULL, move current to current->next and play it.
3. Previous: if current->prev is not NULL, move current to current->prev and play it.

Repeat Playback (Circular Linked List)

1. Add song: if the list is empty, make the node point to itself.
2. Otherwise, set new node->next = last->next, last->next = new node, then last = new node.
3. Repeat play (N songs): start at temp = last->next (first song).
4. Repeat N times: print temp->song, then set temp = temp->next. The pointer loops back to the first song and never reaches NULL.